

CPSC 304 Project Cover Page

Milestone #: 2

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Group Number: 25

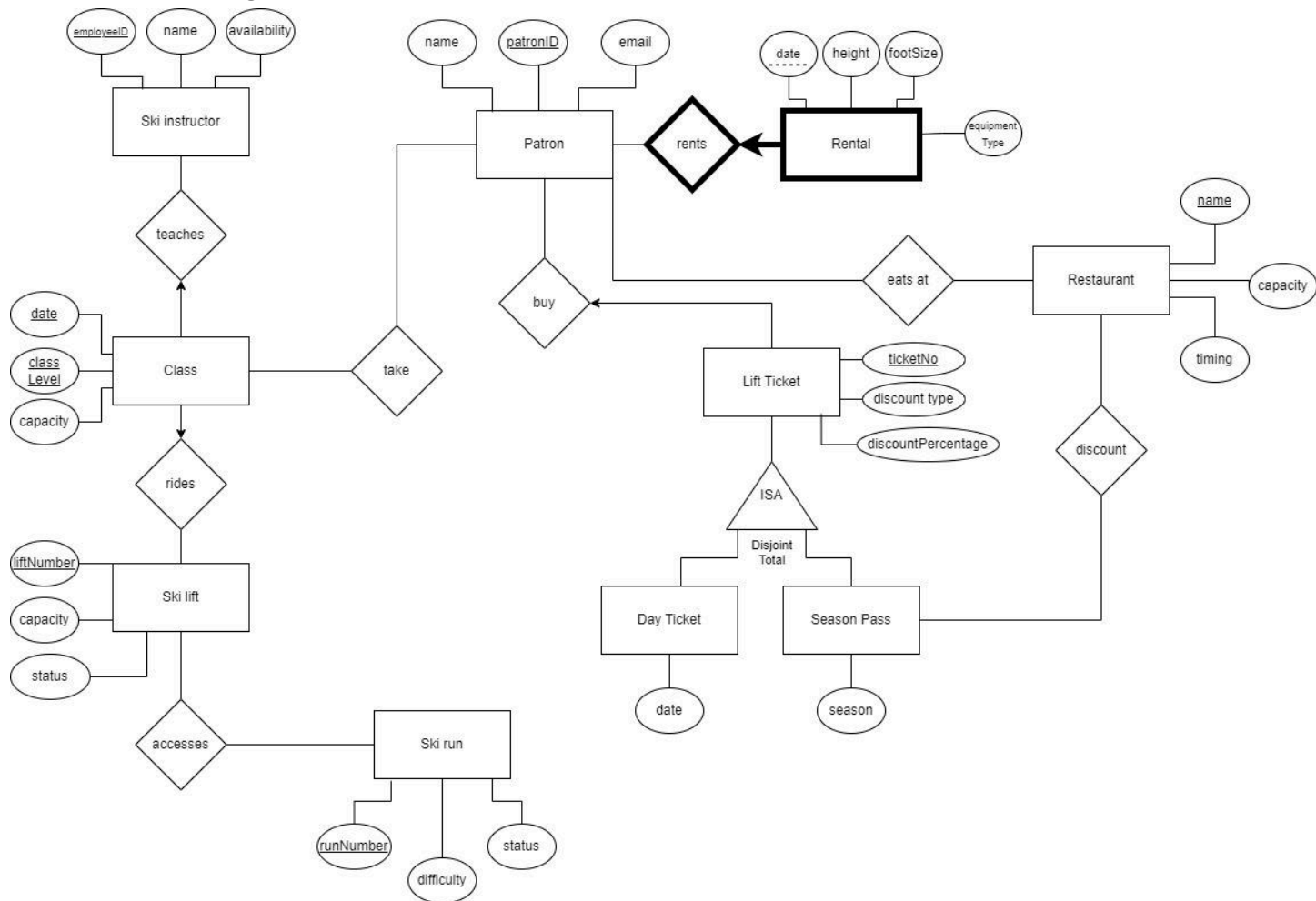
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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia.

Milestone 2:

1. Updated ER Diagram



2. Summary of the Project:

Our project focuses on managing various aspects of a ski resort, including guest management, equipment rentals, lift ticket sales, ski lessons, dining services, and ski infrastructure. The goal is to develop a database that can optimize the operations of the resort by keeping track of all relevant entities and relationships.

3. Changes to the ER Diagram

- Based on the feedback from our TA, we have made the following changes:
 - Patron-Restaurant Relationship:** We have changed the relationship between Patron and Restaurant to a many-to-many relationship to allow patrons to eat at multiple restaurants. This modification is represented by replacing the arrow with a straight line connecting Patron and Restaurant entities. This change was made now to allow for multiple patrons to eat at different restaurants while at the resort.
 - Lift Ticket-Restaurant Relationship:** We have also modified the relationship between Lift Ticket and Restaurant to a many-to-many relationship, indicating that a season pass can allow discounts at multiple restaurants. This change ensures that the system can handle scenarios where season pass holders receive discounts at various restaurants at the resort.

- discountPercent Attribute: We added discountPercent to Lift Ticket to introduce a functional dependency.
- Removed "Access" Relationship: After further discussion, we believe this relationship is unnecessary since any lift ticket gives access to all the lifts and thus making an extra table to manage this relationship would be inefficient.

4. Relational Schema

1. Patron(patronID: VARCHAR(20) PK, name: VARCHAR(50), email: VARCHAR(50) UNIQUE NOT NULL)
2. SkiInstructor(employeeID: VARCHAR(20) PK, name: VARCHAR(50), availability: VARCHAR(50))
3. Class(date: DATE, classLevel: VARCHAR(20), capacity: INT, instructorID: VARCHAR(20) FK, liftNumber: VARCHAR(20) FK, PK(date, classLevel))
4. Take(patronID: VARCHAR(20) PK, date: DATE, classLevel: VARCHAR(20))
5. Rental(date: DATE, height: FLOAT, footSize: FLOAT, equipmentType: VARCHAR(20), patronID: VARCHAR(20) FK, PK(date, patronID))
6. SkiLift(liftNumber: VARCHAR(20) PK, capacity: INT, status: VARCHAR(20))
7. SkiRun(runNumber: VARCHAR(20) PK, difficulty: VARCHAR(20), status: VARCHAR(20))
8. LiftTicket(ticketNo: VARCHAR(20) PK, discountType: VARCHAR(20), discountPercentage: INT, patronID: VARCHAR(20) FK)
9. DayTicket(ticketNo: VARCHAR(20) PK, date: DATE, FK: VARCHAR(20) FK)
10. SeasonPass(ticketNo: VARCHAR(20) PK, season: VARCHAR(20), FK: VARCHAR(20) FK)
11. Restaurant(name: VARCHAR(50), capacity: INT, timing: VARCHAR(50))
12. EatsAt(patronID: VARCHAR(20) FK, restaurantName: VARCHAR(20) FK, PK(patronID, restaurantName))
13. Discount(ticketNo: VARCHAR(20) FK, restaurantName: VARCHAR(20) FK, PK(ticketNo, restaurantName))
14. Accesses(runNumber: VARCHAR(20) FK, liftNumber: VARCHAR(20) FK, PK(runNumber, liftNumber))

5. Functional Dependencies

Patron: patronID -> name, email
email -> patronID, name

SkiInstructor: employeeID -> name, availability

Class: date, classLevel -> capacity, instructorID, liftNumber

Take: patronID -> date, classLevel

Rental: patronID, date -> height, footSize, equipmentType
patronID -> height, footSize

SkiLift: liftNumber -> capacity, status

SkiRun: runNumber -> difficulty, status

LiftTicket: ticketNo -> discountType, discountPercentage, patronID
discountType -> discountPercentage

DayTicket: ticketNo -> date

SeasonPass: ticketNo -> season

Restaurant: restaurantID -> name, capacity

6. Normalization

We will normalize to 3NF.

Patron: Already in 3NF.

Patron(patronID: VARCHAR(20) PK, name: VARCHAR(50), email: VARCHAR(50) UNIQUE NOT NULL)

SkiInstructor: Already in 3NF.

SkiInstructor(employeeID: VARCHAR(20) PK, name: VARCHAR(50), availability: VARCHAR(50))

Class: Already in 3NF.

Class(date: DATE, classLevel: VARCHAR(20), capacity: INT, instructorID: VARCHAR(20) FK, liftNumber: VARCHAR(20) FK, PK(date, classLevel))

Take: Already in 3NF.

Take(patronID: VARCHAR(20) PK, date: DATE, classLevel: VARCHAR(20))

Rental:

Rental(date: DATE, height: FLOAT, footSize: FLOAT, equipmentType: VARCHAR(20), patronID: VARCHAR(20) FK, PK(date, patronID))

Minimal Cover:

patronID, date -> equipmentType (Does not violate 3NF.)

patronID -> height

patronID -> footSize

Decomposition to BCNF:

Rental1(date: DATE, patronID, equipmentType, footSize, PK(patronID, date))

Rental2(patronID PK FK, height)

Rental3(date, patronID, equipmentType, PK(patronID, date))

Rental4(patronID PK FK, footSize)

We keep Rental2, Rental3, and Rental4. All FDs are preserved.

SkiLift: Already in 3NF.

SkiLift(liftNumber: VARCHAR(20) PK, capacity: INT, status: VARCHAR(20))

SkiRun: Already in 3NF.

SkiRun(runNumber: VARCHAR(20) PK, difficulty: VARCHAR(20), status: VARCHAR(20))

LiftTicket:

LiftTicket(ticketNo: VARCHAR(20) PK, discountType: VARCHAR(20), discountPercentage: INT, patronID: VARCHAR(20) FK)

Minimal Cover:

ticketNo -> discountType (Does not violate 3NF.)

ticketNo -> discountPercentage (Does not violate 3NF.)

discountType -> discountPercentage

Decomposition to BCNF:

LiftTicket1(ticketNo: VARCHAR(20) PK, discountType: VARCHAR(20) FK, patronID: VARCHAR(20) FK)

LiftTicket2(discountType VARCHAR(20) PK, discountPercentage: INT)

We keep both LiftTicket1 and LiftTicket2. All FDs are preserved.

DayTicket: Already in 3NF.

DayTicket(ticketNo: VARCHAR(20) PK, date: DATE, FK: VARCHAR(20) FK)

SeasonPass: Already in 3NF.

SeasonPass(ticketNo: VARCHAR(20) PK, season: VARCHAR(20), FK: VARCHAR(20) FK)

Restaurant: Already in 3NF.

Restaurant(name: VARCHAR(50), capacity: INT, timing: VARCHAR(50))

EatsAt: Already in 3NF.

EatsAt(patronID: VARCHAR(20) FK, restaurantName: VARCHAR(20) FK,
PK(patronID,restaurantName))

Discount: Already in 3NF.

Discount(ticketNo: VARCHAR(20) FK, restaurantName: VARCHAR(20) FK, PK(ticketNo, restaurantName))

Accesses: Already in 3NF.

```
Accesses(runNumber: VARCHAR(20) FK, liftNumber: VARCHAR(20) FK, PK(runNumber, liftNumber))
```

7. DDL Statements

```
CREATE TABLE Patron (
    patronID VARCHAR(20) PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50) UNIQUE NOT NULL
);
```

```
CREATE TABLE SkiInstructor (
    employeeID VARCHAR(20) PRIMARY KEY,
    name VARCHAR(50),
    availability VARCHAR(50)
);
```

```
CREATE TABLE Class (
```

```

    date DATE,
    classLevel VARCHAR(20),
    capacity INT,
    instructorID VARCHAR(20),
    liftNumber VARCHAR(20),
    PRIMARY KEY (date, classLevel),
    FOREIGN KEY (instructorID) REFERENCES SkiInstructor(employeeID)
        ON DELETE SET NULL,
    FOREIGN KEY (liftNumber) REFERENCES SkiLift(liftNumber)
        ON DELETE SET NULL
);

CREATE TABLE Take (
    patronID VARCHAR(20),
    date DATE,
    classLevel VARCHAR(20),
    PRIMARY KEY (patronID, date, classLevel),
    FOREIGN KEY (patronID) REFERENCES Patron(patronID)
        ON DELETE CASCADE,
    FOREIGN KEY (date, classLevel) REFERENCES Class(date, classLevel)
        ON DELETE CASCADE
);

CREATE TABLE Rental2 (
    patronID VARCHAR(20) PRIMARY KEY,
    height FLOAT,
    FOREIGN KEY (patronID) REFERENCES Patron(patronID)
        ON DELETE CASCADE
);

CREATE TABLE Rental3 (
    date DATE,
    patronID VARCHAR(20),
    equipmentType VARCHAR(20),
    PRIMARY KEY (patronID, date),
    FOREIGN KEY (patronID) REFERENCES Patron(patronID)
        ON DELETE CASCADE
);

CREATE TABLE Rental4 (

```

```

        patronID VARCHAR(20) PRIMARY KEY,
        footSize FLOAT,
        FOREIGN KEY (patronID) REFERENCES Patron(patronID)
            ON DELETE CASCADE
    );

CREATE TABLE SkiLift (
    liftNumber VARCHAR(20) PRIMARY KEY,
    capacity INT,
    status VARCHAR(20)
);

CREATE TABLE SkiRun (
    runNumber VARCHAR(20) PRIMARY KEY,
    difficulty VARCHAR(20),
    status VARCHAR(20)
);

CREATE TABLE LiftTicket1 (
    ticketNo VARCHAR(20) PRIMARY KEY,
    discountType VARCHAR(20),
    patronID VARCHAR(20),
    FOREIGN KEY (patronID) REFERENCES Patron(patronID)
        ON DELETE CASCADE,
    FOREIGN KEY (discountType) REFERENCES LiftTicket2(discountType)
        ON DELETE CASCADE
);

CREATE TABLE LiftTicket2 (
    discountType VARCHAR(20) PRIMARY KEY,
    discountPercentage INT
);

CREATE TABLE DayTicket (
    ticketNo VARCHAR(20) PRIMARY KEY,
    date DATE,
    FOREIGN KEY (ticketNo) REFERENCES LiftTicket1(ticketNo)
        ON DELETE CASCADE
);

CREATE TABLE SeasonPass (

```

```

        ticketNo VARCHAR(20) PRIMARY KEY,
        season VARCHAR(20),
        FOREIGN KEY (ticketNo) REFERENCES LiftTicket1(ticketNo)
            ON DELETE CASCADE
    );

CREATE TABLE Restaurant (
    name VARCHAR(50) PRIMARY KEY,
    capacity INT,
    timing VARCHAR(50)
);

CREATE TABLE EatsAt (
    patronID VARCHAR(20),
    restaurantName VARCHAR(20),
    PRIMARY KEY (patronID, restaurantName),
    FOREIGN KEY (patronID) REFERENCES Patron(patronID)
        ON DELETE CASCADE,
    FOREIGN KEY (restaurantName) REFERENCES Restaurant(name)
        ON DELETE CASCADE
);

CREATE TABLE Discount (
    ticketNo VARCHAR(20),
    restaurantName VARCHAR(20),
    PRIMARY KEY (ticketNo, restaurantName),
    FOREIGN KEY (ticketNo) REFERENCES LiftTicket1(ticketNo)
        ON DELETE CASCADE,
    FOREIGN KEY (restaurantName) REFERENCES Restaurant(name)
        ON DELETE CASCADE
);

CREATE TABLE Accesses (
    runNumber VARCHAR(20),
    liftNumber VARCHAR(20),
    PRIMARY KEY (runNumber, liftNumber),
    FOREIGN KEY (runNumber) REFERENCES SkiRun(runNumber)
        ON DELETE CASCADE,
    FOREIGN KEY (liftNumber) REFERENCES SkiLift(liftNumber)
        ON DELETE CASCADE
);

```

8. Insert Statements

1. Patron

```
INSERT INTO Patron (patronID, name, email) VALUES ('P001', 'Jake John', 'jjohn2@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P002', 'Lili Henrik', 'lilihenn@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P003', 'Jasper Deasey', 'jasperstezy@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P004', 'Ella Krav', 'ellakrav@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P005', 'Blake Smith', 'smithblake@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P006', 'Emily Johnson', 'emily.j@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P007', 'Sarah Brown', 'sbrown@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P008', 'David Wilson', 'dave.r.wilson@gmail.com');
INSERT INTO Patron (patronID, name, email) VALUES ('P009', 'Jessica Martinez', 'jess.mrtnz@gmail.com');
```

2. Ski Instructor

```
INSERT INTO SkiInstructor (employeeID, name, availability) VALUES ('E001', 'Instructor One', 'Available');
INSERT INTO SkiInstructor (employeeID, name, availability) VALUES ('E002', 'Instructor Two', 'Available');
INSERT INTO SkiInstructor (employeeID, name, availability) VALUES ('E003', 'Instructor Three', 'Available');
INSERT INTO SkiInstructor (employeeID, name, availability) VALUES ('E004', 'Instructor Four', 'Available');
INSERT INTO SkiInstructor (employeeID, name, availability) VALUES ('E005', 'Instructor Five', 'Unavailable');
```

3. Class

```
INSERT INTO Class (date, classLevel, capacity, instructorID, liftNumber) VALUES ('2024-12-20', 'Beginner', 5, 'E001', 'L001');
INSERT INTO Class (date, classLevel, capacity, instructorID, liftNumber) VALUES ('2024-12-20', 'Intermediate', 10, 'E001', 'L002');
```

```
INSERT INTO Class (date, classLevel, capacity, instructorID,
liftNumber) VALUES ('2024-12-21', 'Advanced', 10, 'E001', 'L004');
INSERT INTO Class (date, classLevel, capacity, instructorID,
liftNumber) VALUES ('2024-12-22', 'Beginner', 10, 'E001', 'L001');
INSERT INTO Class (date, classLevel, capacity, instructorID,
liftNumber) VALUES ('2024-12-22', 'Intermediate', 10, 'E001', 'L002');
```

4. Take

```
INSERT INTO Take (patronID, date, classLevel) VALUES ('P001',
'2024-12-20', 'Beginner');
INSERT INTO Take (patronID, date, classLevel) VALUES ('P001',
'2024-07-22', 'Beginner');
INSERT INTO Take (patronID, date, classLevel) VALUES ('P002',
'2024-12-21', 'Advanced');
INSERT INTO Take (patronID, date, classLevel) VALUES ('P004',
'2024-12-20', 'Intermediate');
INSERT INTO Take (patronID, date, classLevel) VALUES ('P005',
'2024-12-22', 'Intermediate');
```

5. Rental

```
INSERT INTO Rental3 (date, patronID, equipmentType) VALUES
('2024-12-20', 'P001', 'Skis+Boots');
INSERT INTO Rental3 (date, patronID, equipmentType) VALUES
('2024-12-22', 'P001', 'Skis+Boots');
INSERT INTO Rental3 (date, patronID, equipmentType) VALUES
('2024-12-20', 'P004', 'Skis+Boots');
INSERT INTO Rental3 (date, patronID, equipmentType) VALUES
('2024-12-22', 'P005', 'Boots');
INSERT INTO Rental3 (date, patronID, equipmentType) VALUES
('2024-12-27', 'P005', 'Boots');
```

```
INSERT INTO Rental2 (patronID, height) VALUES ('P001', 5.8);
INSERT INTO Rental2 (patronID, height) VALUES ('P004', 6.1);
INSERT INTO Rental2 (patronID, height) VALUES ('P005', 5.8);
```

```
INSERT INTO Rental4 (patronID, footSize) VALUES ('P001', 9.5);
INSERT INTO Rental4 (patronID, footSize) VALUES ('P004', 12.0);
INSERT INTO Rental4 (patronID, footSize) VALUES ('P005', 8.0);
```

6. SkiLift

```
INSERT INTO SkiLift (liftNumber, capacity, status) VALUES ('L001', 4,
'Operational');
INSERT INTO SkiLift (liftNumber, capacity, status) VALUES ('L002', 6,
'Operational');
INSERT INTO SkiLift (liftNumber, capacity, status) VALUES ('L003', 4,
'Under Maintenance');
INSERT INTO SkiLift (liftNumber, capacity, status) VALUES ('L004', 4,
'Operational');
INSERT INTO SkiLift (liftNumber, capacity, status) VALUES ('L005', 6,
'Operational');
```

7. SkiRun

```
INSERT INTO SkiRun (runNumber, difficulty, status) VALUES ('R001',
'Green', 'Open');
INSERT INTO SkiRun (runNumber, difficulty, status) VALUES ('R002',
'Green', 'Open');
INSERT INTO SkiRun (runNumber, difficulty, status) VALUES ('R003',
'Black', 'Closed');
INSERT INTO SkiRun (runNumber, difficulty, status) VALUES ('R004',
'Double Black', 'Open');
INSERT INTO SkiRun (runNumber, difficulty, status) VALUES ('R005',
'Blue', 'Open');
```

8. LiftTicket

```
INSERT INTO LiftTicket2 (discountType, discountPercentage) VALUES
('None', 0);
INSERT INTO LiftTicket2 (discountType, discountPercentage) VALUES
('Student', 25);
INSERT INTO LiftTicket2 (discountType, discountPercentage) VALUES
('Senior', 30);
```

```
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('D001', 'None', 'P001', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('D002', 'None', 'P002', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('D003', 'Student', 'P003', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('D004', 'None', 'P004', NULL);
```

```
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('D005', 'Senior', 'P005', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('S001', 'Student', 'P006', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('S002', 'None', 'P007', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('S003', 'Senior', 'P008', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('S004', 'None', 'P009', NULL);
INSERT INTO LiftTicket1 (ticketNo, discountType, patronID, liftNumber)
VALUES ('S005', 'None', 'P004', NULL);
```

9. DayTicket

```
INSERT INTO DayTicket (ticketNo, date) VALUES ('D001', '2024-12-20');
INSERT INTO DayTicket (ticketNo, date) VALUES ('D002', '2024-12-21');
INSERT INTO DayTicket (ticketNo, date) VALUES ('D003', '2024-12-22');
INSERT INTO DayTicket (ticketNo, date) VALUES ('D004', '2024-12-20');
INSERT INTO DayTicket (ticketNo, date) VALUES ('D005', '2024-12-21');
```

10. SeasonPass

```
INSERT INTO SeasonPass (ticketNo, season) VALUES ('S001',
'2024-2025');
INSERT INTO SeasonPass (ticketNo, season) VALUES ('S002',
'2024-2025');
INSERT INTO SeasonPass (ticketNo, season) VALUES ('S003',
'2024-2025');
INSERT INTO SeasonPass (ticketNo, season) VALUES ('S004',
'2023-2024');
INSERT INTO SeasonPass (ticketNo, season) VALUES ('S005',
'2023-2024');
```

11. Restaurant

```
INSERT INTO Restaurant (restaurantID, name, capacity, timing) VALUES
('R001', 'Mountain Top', 100, '09:00-21:00');
INSERT INTO Restaurant (restaurantID, name, capacity, timing) VALUES
('R002', 'Creekside', 75, '11:00-20:00');
INSERT INTO Restaurant (restaurantID, name, capacity, timing) VALUES
('R003', 'Glacier Hut', 50, '09:00-15:00');
```

```
INSERT INTO Restaurant (restaurantID, name, capacity, timing) VALUES  
( 'R004', 'Peak View', 50, '09:00-15:00');  
INSERT INTO Restaurant (restaurantID, name, capacity, timing) VALUES  
( 'R005', 'Round House', 250, '07:00-17:00');
```

12. EatsAt

```
INSERT INTO EatsAt (patronID, restaurantID) VALUES ( 'P002', 'R005');  
INSERT INTO EatsAt (patronID, restaurantID) VALUES ( 'P003', 'R002');  
INSERT INTO EatsAt (patronID, restaurantID) VALUES ( 'P004', 'R001');  
INSERT INTO EatsAt (patronID, restaurantID) VALUES ( 'P005', 'R002');  
INSERT INTO EatsAt (patronID, restaurantID) VALUES ( 'P003', 'R003');
```

13. Discount

```
INSERT INTO Discount (ticketNo, restaurantID) VALUES ( 'S002', 'R001');  
INSERT INTO Discount (ticketNo, restaurantID) VALUES ( 'S002', 'R002');  
INSERT INTO Discount (ticketNo, restaurantID) VALUES ( 'S002', 'R003');  
INSERT INTO Discount (ticketNo, restaurantID) VALUES ( 'S003', 'R001');  
INSERT INTO Discount (ticketNo, restaurantID) VALUES ( 'S003', 'R002');  
INSERT INTO Discount (ticketNo, restaurantID) VALUES ( 'S003', 'R003');
```

14. Accesses

```
INSERT INTO Accesses (runNumber, liftNumber) VALUES ( 'R001', 'L001');  
INSERT INTO Accesses (runNumber, liftNumber) VALUES ( 'R002', 'L002');  
INSERT INTO Accesses (runNumber, liftNumber) VALUES ( 'R003', 'L003');  
INSERT INTO Accesses (runNumber, liftNumber) VALUES ( 'R004', 'L004');  
INSERT INTO Accesses (runNumber, liftNumber) VALUES ( 'R005', 'L005');
```